

IT-SIMPLERA Institute



Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT

Computer Networking Fundamentals & Network Configuration

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Introduction

Computer networking is the foundation of modern communication systems, enabling computers and devices to exchange data efficiently over local and global networks. Understanding networking concepts is essential for cybersecurity professionals because secure communication depends on properly configured and managed networks.

This assignment focuses on identifying network configuration details, using basic networking commands, comparing networking devices and Internet Protocol versions, and understanding the OSI Model. These activities help develop practical troubleshooting skills and strengthen the theoretical knowledge required in cybersecurity.

Task 1: Network Information

1. IPv4 Address: 192.168.199.91

An IPv4 address is a unique numerical identifier assigned to a device on a network. It enables the device to communicate with other devices and access network resources using the Internet Protocol (IPv4).

2. MAC Address: 74-D8-3E-F0-18-C9

A MAC (Media Access Control) address is a unique hardware address assigned to a network interface card (NIC). It is used to identify a device on a local area network (LAN).

3. Default Gateway: 192.168.199.223

The Default Gateway is the network device (usually a router) that forwards data from the local network to other networks, including the Internet. It serves as the exit point for network traffic.

```
C:\WINDOWS\system32\cmd. x + v
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 74-D8-3E-F0-18-C9
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2c8f:f5c0:600:1351::101(Preferred)
Lease Obtained. . . . . : Saturday, July 11, 2026 1:58:32 PM
Lease Expires . . . . . : Sunday, July 12, 2026 1:58:31 PM
IPv6 Address. . . . . : 2c8f:f5c0:600:1351::d8bc:48a3:adcf:5b13(Preferred)
Temporary IPv6 Address. . . . . : 2c8f:f5c0:600:1351::35a5:7d52:7476:3401(Preferred)
Link-local IPv6 Address . . . . . : fe80::9fb:3245:ab63:ff8c%9(Preferred)
IPv4 Address. . . . . : 192.168.199.91(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, July 11, 2026 1:58:33 PM
Lease Expires . . . . . : Saturday, July 11, 2026 3:28:38 PM
Default Gateway . . . . . : fe80::50c5:9cff:fe7f:6dd1%9
                          192.168.199.223
DHCP Server . . . . . : 192.168.199.223
DHCPv6 IAID . . . . . : 141875262
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-AA-61-74-D8-3E-F0-18-C9
DNS Servers . . . . . : 240c::6666
                          240c::6644
                          192.168.199.223
                          2c8f:f5c0:600:1351::6a
                          240c::6666
                          240c::6644
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 74-D8-3E-F0-18-CD
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\hpx>

C:\WINDOWS\system32\cmd. x + v
Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
Physical Address. . . . . : 08-50-56-C0-00-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::fc71:3d3a:819e:20c5%11(Preferred)
IPv4 Address. . . . . : 192.168.85.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, July 11, 2026 1:57:32 PM
Lease Expires . . . . . : Saturday, July 11, 2026 3:12:54 PM
Default Gateway . . . . . : 
DHCP Server . . . . . : 192.168.85.254
DHCPv6 IAID . . . . . : 738218070
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-AA-61-A1-74-D8-3E-F0-18-C9
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 08-50-56-C0-00-08
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::db83:949d:e5ba:1119%3(Preferred)
IPv4 Address. . . . . : 192.168.30.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, July 11, 2026 1:57:24 PM
Lease Expires . . . . . : Saturday, July 11, 2026 3:12:55 PM
Default Gateway . . . . . : 
DHCP Server . . . . . : 192.168.30.254
DHCPv6 IAID . . . . . : 754995286
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-AA-61-A1-74-D8-3E-F0-18-C9
Primary WINS Server . . . . . : 192.168.30.2
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
```

Task 2: Basic Networking Commands

```
C:\WINDOWS\system32\cmd. x + v

C:\Users\hp>ping google.com

Pinging google.com [2a00:1450:4006:817::200e] with 32 bytes of data:
Reply from 2a00:1450:4006:817::200e: time=129ms
Reply from 2a00:1450:4006:817::200e: time=164ms
Reply from 2a00:1450:4006:817::200e: time=130ms
Reply from 2a00:1450:4006:817::200e: time=122ms

Ping statistics for 2a00:1450:4006:817::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 122ms, Maximum = 164ms, Average = 138ms

C:\Users\hp>tracert google.com

Tracing route to google.com [2a00:1450:4006:817::200e]
over a maximum of 30 hops:

  1    24 ms    3 ms    2c0f:f5c0:71f:242::43
  2   120 ms   46 ms   160 ms  2c0f:f5c0:71f:242::43 [2a00:1450:4006:817::200e]

Trace complete.

C:\Users\hp>nslookup google.com
Server:      Unknown
Address:     240c::6666

Non-authoritative answer:
DNS request timed out.
    timeout was 2 seconds.
Name:       google.com
Address:    142.251.209.78

C:\Users\hp>arp -a

Interface: 192.168.30.1 --- 0x3
Internet Address      Physical Address      Type
192.168.30.254        00-50-56-e6-67-c3    dynamic
192.168.30.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

```
C:\Users\hp>arp -a

Interface: 192.168.30.1 --- 0x3
Internet Address      Physical Address      Type
192.168.30.254        00-50-56-e6-67-c3    dynamic
192.168.30.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.199.91 --- 0x9
Internet Address      Physical Address      Type
192.168.199.223       e2-01-7e-b4-fc-ec    dynamic
192.168.199.255       ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.85.1 --- 0xb
Internet Address      Physical Address      Type
192.168.85.254        00-50-56-ec-0a-26    dynamic
192.168.85.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0xd
Internet Address      Physical Address      Type
192.168.56.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
```

Command

Purpose

Displays detailed network configuration information, including the computer's IP address, MAC address, DNS servers, and default gateway. It is used to troubleshoot network connectivity issues.

Command	Purpose
<code>ping google.com</code>	Tests connectivity between the computer and a remote host by sending ICMP Echo Request packets. It also measures the response time and checks for packet loss.
<code>tracert google.com</code>	Displays the path (route) that packets take from your computer to the destination host. It helps identify where delays or network problems occur.
<code>nslookup google.com</code>	Queries the Domain Name System (DNS) to translate a domain name into its corresponding IP address. It is commonly used to verify DNS resolution.
<code>arp -a</code>	Displays the Address Resolution Protocol (ARP) cache, showing the mapping between IP addresses and MAC addresses on the local network. It helps troubleshoot local network communication.

Task 3: Router vs. Switch

Feature	Router	Switch
Primary Function	Connects different networks and forwards data between them.	Connects devices within the same local area network (LAN).
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer) (<i>some advanced switches can operate at Layer 3</i>)
Address Used	Uses IP addresses to route packets.	Uses MAC addresses to forward frames.
Coverage	Connects multiple networks, such as a home network to the Internet.	Connects multiple devices within a single network.
Broadcast Domain	Separates broadcast domains, reducing unnecessary network traffic.	Devices connected to the same switch are typically in the same broadcast domain unless VLANs are configured.
Common Use Cases	Home routers, office gateways, Internet Service Provider (ISP) networks.	Connecting computers, printers, servers, and access points in offices or schools.

Conclusion

A **router** is used to connect different networks and direct data between them using IP addresses, while a **switch** connects devices within the same network using MAC addresses. Both are essential networking devices but serve different purposes.

Task 4: IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32-bit address	128-bit address
Address Format	Four decimal numbers separated by dots	Eight groups of hexadecimal numbers separated by colons
Example Address	192.168.199.91	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Number of Addresses	Approximately 4.3 billion addresses	Approximately 340 undecillion (3.4×10^{38}) addresses
Configuration	Can be configured manually or using DHCP	Supports automatic configuration (SLAAC) and DHCPv6
Security	IPsec is optional	IPsec support is built into the protocol (though its use is optional)
Broadcast Support	Supports broadcast, multicast, and unicast	Does not support broadcast; uses multicast and anycast instead
Header Size	Variable header (20–60 bytes)	Fixed header (40 bytes), making packet processing more efficient

Conclusion

IPv4 is the older Internet Protocol with a limited address space, while IPv6 is the newer version designed to provide a vastly larger number of addresses, improved efficiency, and better support for modern networking requirements.

Task 5: OSI Model

Layer (Top to Bottom)	Primary Function	Common Protocol	Networking Device
7. Application	Provides network services directly to end-user applications.	HTTP	Proxy Server
6. Presentation	Translates, encrypts, and compresses data between the application and network.	SSL/TLS	Gateway
5. Session	Establishes, manages, and terminates communication sessions between devices.	NetBIOS	Gateway
4. Transport	Ensures reliable data delivery, error checking, and flow control.	TCP	Firewall

Layer (Top to Bottom)	Primary Function	Common Protocol	Networking Device
3. Network	Determines the best path for data and routes packets between networks.	IP (IPv4/IPv6)	Router
2. Data Link	Transfers data between devices on the same network and uses MAC addresses.	Ethernet (IEEE 802.3)	Switch
1. Physical	Transmits raw bits over the physical transmission medium.	Ethernet (Physical Layer Standard)	Hub/Cables/Repeater

OSI Model Summary

- **Layer 7 – Application:** Provides services for user applications such as web browsing and email.
- **Layer 6 – Presentation:** Formats, encrypts, and compresses data.
- **Layer 5 – Session:** Creates, maintains, and closes communication sessions.
- **Layer 4 – Transport:** Ensures reliable and complete data transfer using protocols such as TCP.
- **Layer 3 – Network:** Routes packets between different networks using IP addresses.
- **Layer 2 – Data Link:** Delivers frames within the same network using MAC addresses.
- **Layer 1 – Physical:** Transmits binary data through physical media such as cables or wireless signals.

Challenges Faced

During this assignment, I encountered a few challenges while identifying the correct network adapter because my computer contained multiple VMware virtual network adapters in addition to the Wi-Fi adapter. I also learned how to interpret the output of networking commands such as **ipconfig**, **ping**, **tracert**, **nslookup**, and **arp**. Completing these tasks improved my understanding of network configuration and troubleshooting techniques.

Conclusion

This assignment provided valuable practical experience in computer networking fundamentals. I successfully identified important network configuration details, executed essential networking commands, compared routers and switches, distinguished between IPv4 and IPv6, and studied the seven layers of the OSI Model. The knowledge gained from this assignment has strengthened my networking skills and established a solid foundation for future cybersecurity and network administration tasks.

